

JING LAN

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EDUCATION

University of Wisconsin-Madison <i>Ph.D.</i> in Computer Sciences	2024 – present
The Chinese University of Hong Kong <i>M.Phil.</i> in Computer Science and Engineering	2022 – 2024
Sun Yat-sen University <i>B.Eng.</i> in Computer Science, GPA: 3.9 / 4.0	2018 – 2022

TECHNICAL SKILLS

Infrastructure: Kubernetes, Docker, Ansible, Helm
Cloud / Development: Kubespray, DevSpace, Supabase, Git, CI/CD, React, FastAPI
Systems: Linux, HPC, distributed systems, networking, container orchestration
Languages: Python, C/C++, JavaScript, Shell

RESEARCH & ENGINEERING EXPERIENCES

EFabric: Cloud-Native AI4Science Computing Platform NetLab | 2025 – Present

Public platform: efabric.cs.wisc.edu

Bare-metal Kubernetes platform for scientific computing, providing multi-tenant compute, convenient artifact management, secure cluster access, and petabyte-scale shared storage for domain researchers.

- Architected and deployed the platform across **66 bare-metal servers** (scaling toward **100+ nodes**), owning the full lifecycle from hardware bring-up and cluster provisioning to service delivery and production operations.
- Built **reproducible GPU cluster provisioning** with Ansible and Kubespray, automating Kubernetes setup and scaling, firewall policy enforcement, NVIDIA GPU and monitoring software stacks, and Kueue configuration to accelerate node onboarding and minimize configuration drift.
- Established a **DevSpace-based development and deployment workflow**; built a React frontend with BFF routing, a stateless FastAPI backend for experiment bootstrapping and management, MCP-based agent tools, and an SSH gateway for secure remote worker access.
- Designed and operated a **1.2 PB shared storage platform** with TiKV, SeaweedFS, and JuiceFS, exposing S3 and POSIX interfaces to cluster services and research workloads.
- Implemented **authentication, permission controls, and resource admission** with Supabase; integrated Kueue for queue-based resource partitioning and multi-tenant scheduling. Built **cluster-wide observability** with Prometheus and Grafana covering clusters, servers, storage, and experiments.

Flavius: Disaggregated Graph OLAP Engine Husky Data Lab | 2022 – 2024

- Co-designed a **hybrid pull/push execution pipeline** for Flavius's disaggregated master/compute architecture, using staged dataflow to support complex graph queries with branching and iterative structures, while accelerating per-stage data processing with fine-grained parallel tasks.
- Designed and implemented an **adaptive two-level task scheduler** for concurrent graph queries, using batch size to control per-query resource consumption and task size to tune execution parallelism, balancing **throughput** and **P99 latency** under highly skewed mixed workloads.
- Built a **workload-aware scheduler tuner** using derivative-free coordinate descent and integrated it into a **C++ distributed query engine with 41K LOC**; Evaluated against TigerGraph and Nebula-Graph, Flavius achieved the best P99 latency across LDBC and Amazon workloads, and up to **3.6× lower tail latency** than using fixed scheduling configurations.

Student Cluster Competitions (SCC) Sun Yat-sen University | 2020 – 2022

- Optimized highly parallel HPC workloads across **heterogeneous platforms** including Tianhe-2, UofT Niagara, and Chameleon Cloud, with hands-on experience in cluster deployment, benchmarking, and performance tuning.
- Led the university HPC competition team from 2021 to 2022, organizing weekly technical sessions and practices on cluster management, scientific workload deployment, benchmarking, and **CUDA/OpenMP/MPI** optimization.
- Champion, IndySCC@ACM Supercomputing 2021; Honorable Mention, SCC@ISC High Performance 2021.